Sensitivity Analysis: Influence of Inflow Discharge on Water Depth and Flow Velocity in the Sandey Floodplain

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1 Introduction

The Sandey floodplain near Innertkirchen in the Canton of Bern is formed by the Gadmerwasser and is classified as a floodplain of national importance. During high precipitation events, the river floods the surrounding floodplain. As part of a research project at the ZHAW, a hydrodynamic model of the floodplain was set up using the simulation software BASEMENT. In this exercise, the existing model is used to investigate the following research question: *How does varying the inflow discharge influence the mean value and spatial distribution of water depths and flow velocities?*

2 Methods

The hydrodynamic simulations were carried out with BASEMENT v4.2.0 using a 2D shallow water equation solver. All simulations start from dry initial conditions and run for 7 200 s (2 h) with output stored every 300 s. Only the inflow discharge of the Gadmerwasser was varied between the five scenarios listed in Table 2-1.

R (v4.3.3) was used to configure and orchestrate the simulation runs by modifying the JSON control files. The resulting water depth and velocity fields were extracted from the HDF5 output, linked to the computational mesh and rasterised at 0.5 m resolution. ParaView was used to produce the spatial maps. Density distributions of water depth and flow velocity were generated with the R package `ggplot2`, considering only wetted cells ($h > 0$).

Tab. 2-1: Simulation protocol: control parameters and outcomes for all five discharge scenarios.

		Sim 1	**Sim 2**	**Sim 3**	**Sim 4**	**Sim 5**
Control	Mesh	coarse	coarse	coarse	coarse	coarse
	K_{st}	$1 \times K_{st}$	$1 \times K_{st}$	$1 \times K_{st}$	$1 \times K_{st}$	$1 \times K_{st}$
	Discharge	$1.9\,\mathrm{m^3/s}$	$3.5\,\mathrm{m^3/s}$	$5\,\mathrm{m^3/s}$	$15\,\mathrm{m^3/s}$	$25\,\mathrm{m^3/s}$
Outcome	Duration [s]	198.3	225.6	232.6	334.7	411.6
	Mean depth [m]	0.14	0.18	0.19	0.21	0.20
	Mean velocity [m/s]	0.87	1.10	1.18	1.12	1.08

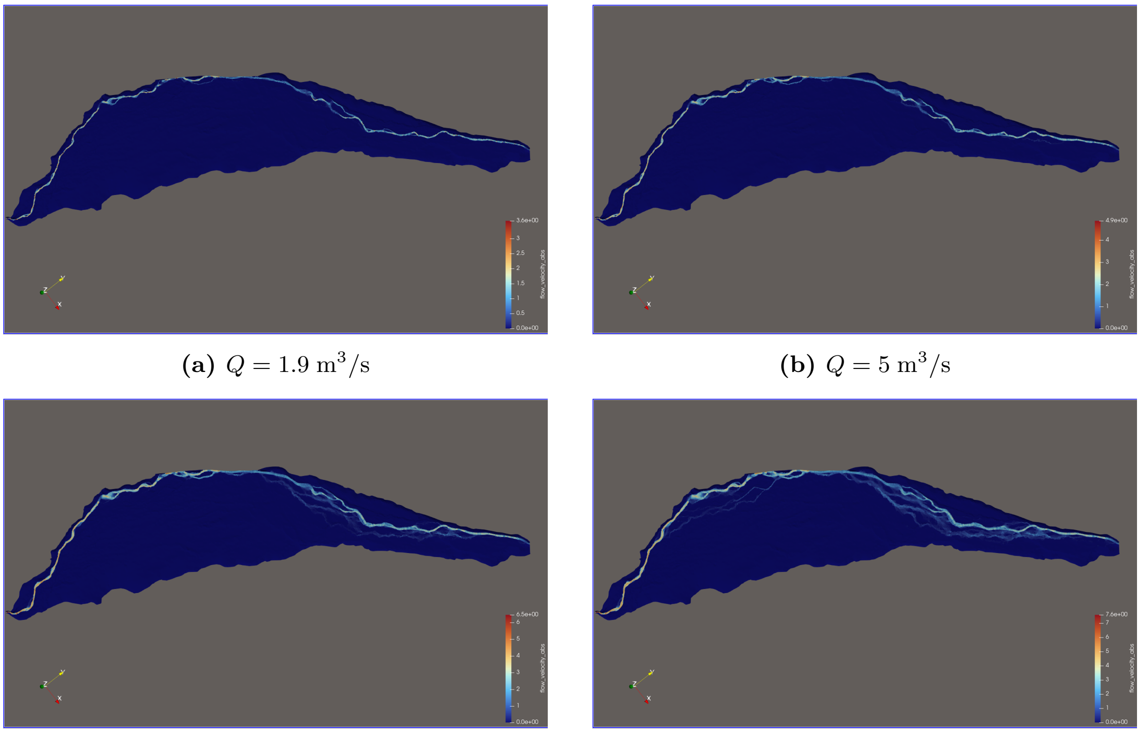
3 Results and Discussion

3.1 Simulation Protocol

Table 2-1 summarises the control parameters and outcomes for all five simulations. The computation time increases with discharge, rising from about 198 s for $Q = 1.9\,\mathrm{m^3/s}$ to roughly 412 s for $Q = 25\,\mathrm{m^3/s}$. The mean water depth increases from 0.14 m to about 0.20 m, while the mean flow velocity peaks at 1.18 m/s for $Q = 5\,\mathrm{m^3/s}$.

3.2 Flow Velocity Maps

The spatial distribution of flow velocities is shown in Figure 3-1. At the lowest discharge ($Q = 1.9\,\mathrm{m^3/s}$), flow is confined to the main channel of the Gadmerwasser, with only minor velocities visible elsewhere. As the discharge increases, more and more side channels across the floodplain become active and carry water.



(a) $Q = 1.9\ \mathrm{m^3/s}$

(b) $Q = 5\ \mathrm{m^3/s}$

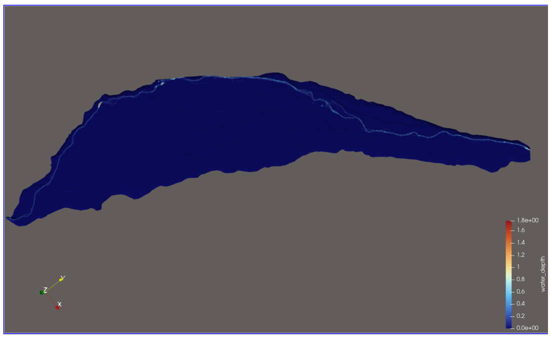
(c) $Q = 15\ \mathrm{m^3/s}$

(d) $Q = 25\ \mathrm{m^3/s}$

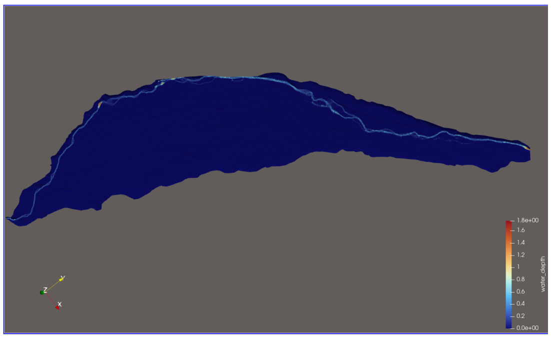
Abb. 3-1: Spatial distribution of absolute flow velocity for four discharge scenarios.

3.3 Water Depth Maps

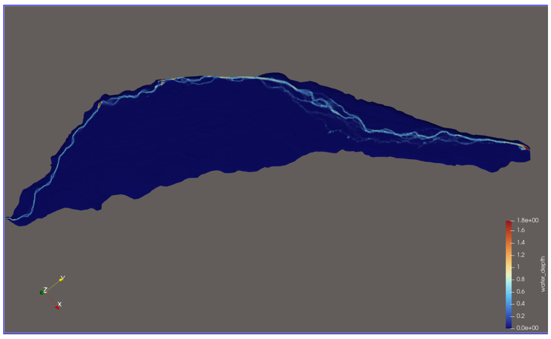
The water depth maps (Figure 3-2) show a very similar pattern. At low discharge, water depth is concentrated in the main channel. With increasing discharge, the side channels fill up and the wetted area expands considerably. The depth values in the main channel also increase noticeably at higher discharges.



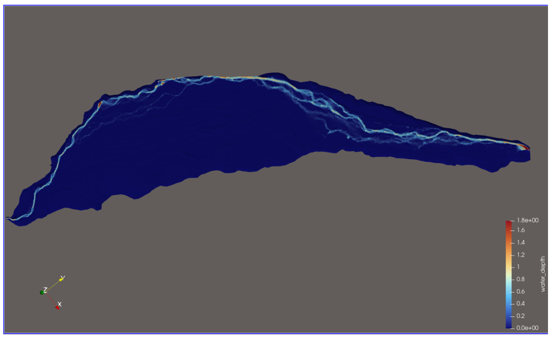
(a) $Q = 1.9\ \mathrm{m}^3/\mathrm{s}$



(b) $Q = 5\ \mathrm{m}^3/\mathrm{s}$



(c) $Q = 15\ \mathrm{m}^3/\mathrm{s}$



(d) $Q = 25\ \mathrm{m}^3/\mathrm{s}$

Abb. 3-2: Spatial distribution of water depth for four discharge scenarios.

3.4 Velocity Distribution

Figure 3-3 shows the density distribution of flow velocities across all wetted cells for each discharge scenario. At higher discharges, the proportion of very low velocity cells increases. This might be because the newly activated side channels tend to carry slow-moving, shallow water. At the same time, the tail of the distribution extends further to the right, suggesting that the main channel reaches higher peak velocities. As a result, the proportion of cells with medium velocities decreases.

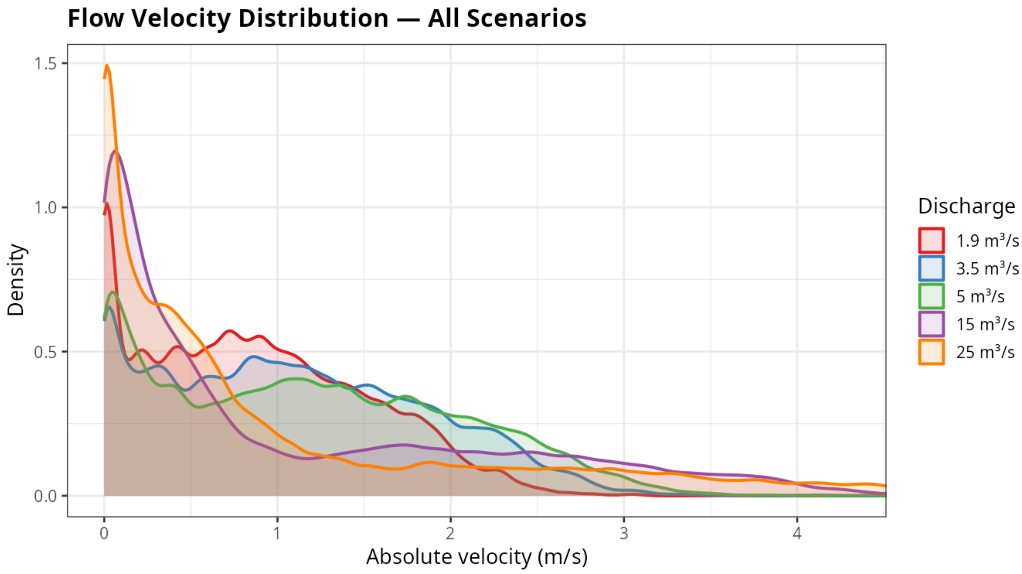


Abb. 3-3: Density distribution of absolute flow velocity across all discharge scenarios. Only wetted cells ($h > 0$) are included.

3.5 Water Depth Distribution

The water depth distribution (Figure 3-4) follows a similar trend. With increasing discharge, the proportion of very shallow cells grows, likely because the newly flooded side channels are generally shallow. At the same time, the water level in the main channel rises, leading to more cells with high depth values. The proportion of cells in the medium depth range decreases accordingly.

This pattern might be explained by two effects acting together: (1) the main channel of the Gadmerwasser carries more water at higher discharges, which increases both depth and velocity in the channel, and (2) the side channels that become active at higher discharges are typically shallow with low to moderate velocities. The combination of these two effects produces the characteristic shift in the distributions, where low and high values increase at the expense of the medium range.

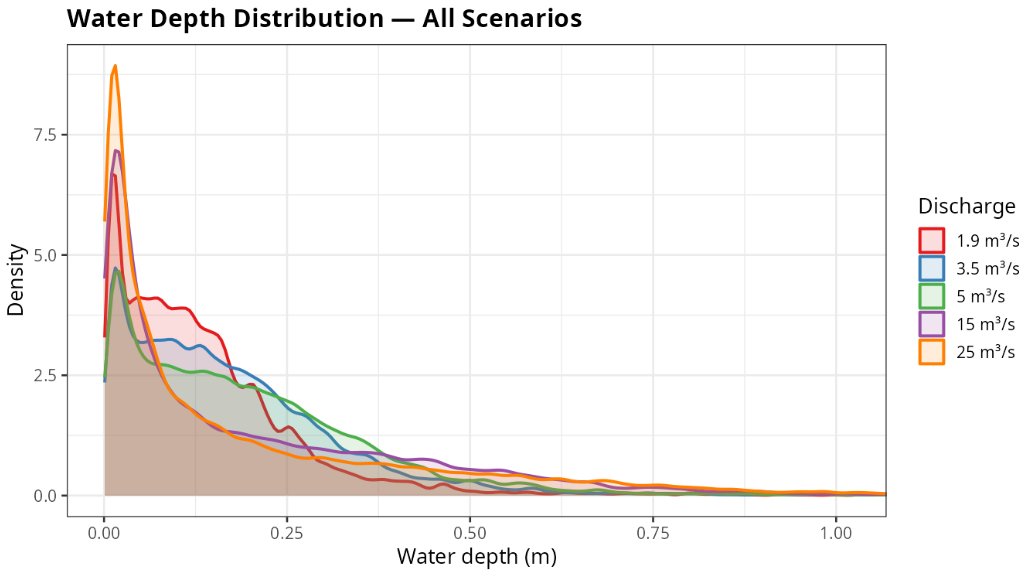


Abb. 3-4: Density distribution of water depth across all discharge scenarios. Only wetted cells ($h > 0$) are included.

3.6 Usage of KI

Claude Opus 4.6 was used as an aid to ensure proper formulation, spelling and grammatical correctness. Further it was used to derive the LaTeX for Images and simulation protocol table.